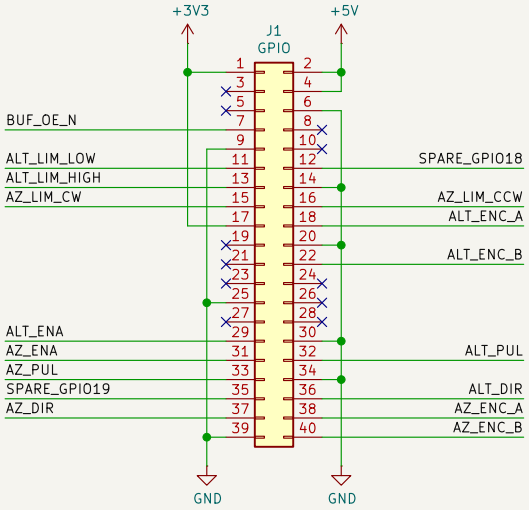




PIN MAP v1, 2026-10-31 (Raspberry Pi 4, BCM GPIO numbers, J1 = 40-pin header)

DRIVER SIGNALS: Pi GPIO → 74ACT244 → DM542T minus pins (common anode)
ALT_PUL GPIO12 pin 32 → 1A1 (2) / 1Y1 (18) → ALT PUL- [hardware PWM0]
ALT_DIR GPIO16 pin 36 → 1A2 (4) / 1Y2 (16) → ALT DIR-
ALT_ENA GPIO5 pin 29 → 1A3 (6) / 1Y3 (14) → ALT ENA-
AZ_PUL GPIO13 pin 33 → 2A1 (11) / 2Y1 (9) → AZ PUL- [hardware PWM1]
AZ_DIR GPIO26 pin 37 → 2A2 (13) / 2Y2 (7) → AZ DIR-
AZ_ENA GPIO6 pin 31 → 2A3 (15) / 2Y3 (5) → AZ ENA-
BUF_OE_N GPIO4 pin 7 → 1OE (1) + 2OE (19), 10k pull-up to +3V3
Unused 244 inputs 1A4 (8), 2A4 (17) → GND. VCC (20) = +5V with 100n to GND (10).
+5V → PUL+, DIR+, ENA+ of both drivers (same +5V as the 244). DM542T logic selector → 5V (ships at 24V).

LIMIT SWITCHES: NC contact between net and GND, 10k pull-up to +3V3, 100n to GND
ALT_LIM_LOW GPIO17 pin 11 | ALT_LIM_HIGH GPIO27 pin 13
AZ_LIM_CW GPIO22 pin 15 | AZ_LIM_CCW GPIO23 pin 16
Reads LOW normally. HIGH = switch hit OR wire broken → software stops that axis.

ENCODERS (none bought yet, spots only): 3.3V max into the Pi
ALT_ENC_A GPIO24 pin 18 | ALT_ENC_B GPIO25 pin 22
AZ_ENC_A GPIO20 pin 38 | AZ_ENC_B GPIO21 pin 40

SOFTWARE LOGIC (common anode flips it: GPIO LOW = optocoupler ON)
PUL: idle HIGH. A step happens on HIGH → LOW. Low ≥ 2.5 µs, high ≥ 2.5 µs (200 kHz max).
DIR: set ≥ 5 µs before the step edge. Which level is clockwise gets set at the bench test.
ENA: HIGH = driver ON, holding torque (same as ENA unwired). LOW = driver OFF, motor turns by hand.
BUF_OE_N: HIGH (boot default, pulled up) = 244 outputs off: drivers ON and holding, no steps.
Software sets PUL HIGH, ENA, DIR first, then drives BUF_OE_N LOW.

LEFT FREE (X on J1): pins 27/28 ID EEPROM only (HAT spec). GPIO2/3 I2C1 (1.8k pull-ups on the Pi) for a future sensor. GPIO14/15 UART for serial console / GPS. GPIO7-11 SPI0 for SPI absolute encoders.
SPARE: GPIO18 (pin 12), GPIO19 (pin 35).

J1 net labels = the Pi pin map. Drop-in for the HAT signal-side schematic.

MV Astronomy

Sheet: /
File: mv-telescope-hat.kicad_sch

Title: MV Telescope Pi HAT: pin map

Size: A4	Date: 2026-10-31	**Rev: pin map v1**
KiCad E.D.A. 10.0.7		Id: 1/1